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9702/52

October/November 2012

1 hour 15 minutes

No Additional Materials are required.

Write your Centre number, candidate number and name on all the work you hand in.
Write in dark blue or black pen.
You may use a soft pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.
DO NOT WRITE IN ANY BARCODES.

You may lose marks if you do not show your working or if you do not use appropriate units.

At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use	
1	
2	
Total	

This document consists of **8** printed pages.

- 1 As a bar magnet is dropped through a coil, an e.m.f. is induced in the coil. The maximum e.m.f. E is induced as the magnet leaves the coil with speed v .

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It is suggested that E is directly proportional to v .

Design a laboratory experiment to test the relationship between E and v . You should draw a diagram, on page 3, showing the arrangement of your equipment. In your account you should pay particular attention to

- (a) the procedure to be followed,
- (b) the measurements to be taken,
- (c) the control of variables,
- (d) the analysis of the data,
- (e) the safety precautions to be taken.

[15]

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For Examiner's Use	Defining the problem	Methods of data collection	Method of analysis	Safety considerations	Additional detail

- 2 Different light-emitting diodes (LEDs) can emit electromagnetic radiation of different wavelengths.

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A student uses the circuit of Fig. 2.1 to investigate the minimum potential difference required to cause an LED to emit its characteristic wavelength.

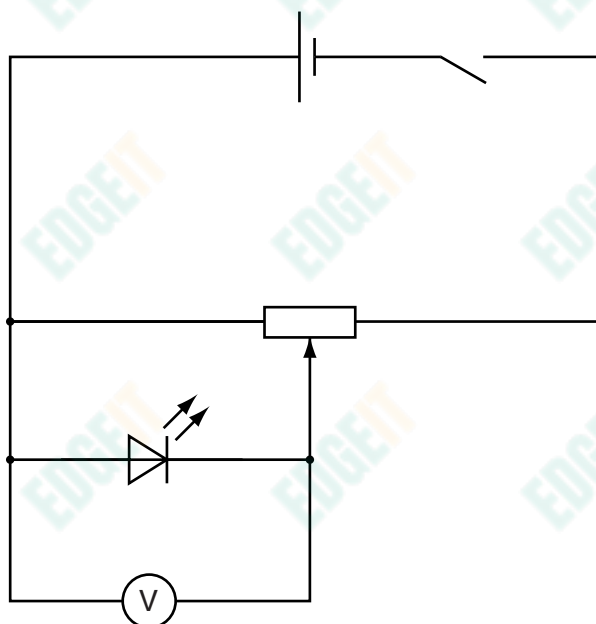


Fig. 2.1

For different LEDs emitting radiation of wavelength λ , the minimum potential difference V is recorded.

Question 2 continues on the next page.

It is suggested that V and λ are related by the equation

$$\frac{hc}{\lambda} = B + eV$$

where c is the speed of light in a vacuum, e is the elementary charge, h is the Planck constant and B is a constant.

- (a) A graph is plotted of V on the y -axis against $1/\lambda$ on the x -axis. Determine expressions for the gradient and y -intercept in terms of B , c , e and h .

gradient =

y -intercept =

[1]

- (b) Values of λ and V are given in Fig. 2.2.

$\lambda/10^{-9}\text{m}$	V/V	
950	0.60 ± 0.05	
875	0.70 ± 0.05	
655	1.20 ± 0.05	
560	1.55 ± 0.05	
505	1.80 ± 0.05	
430	2.25 ± 0.05	

Fig. 2.2

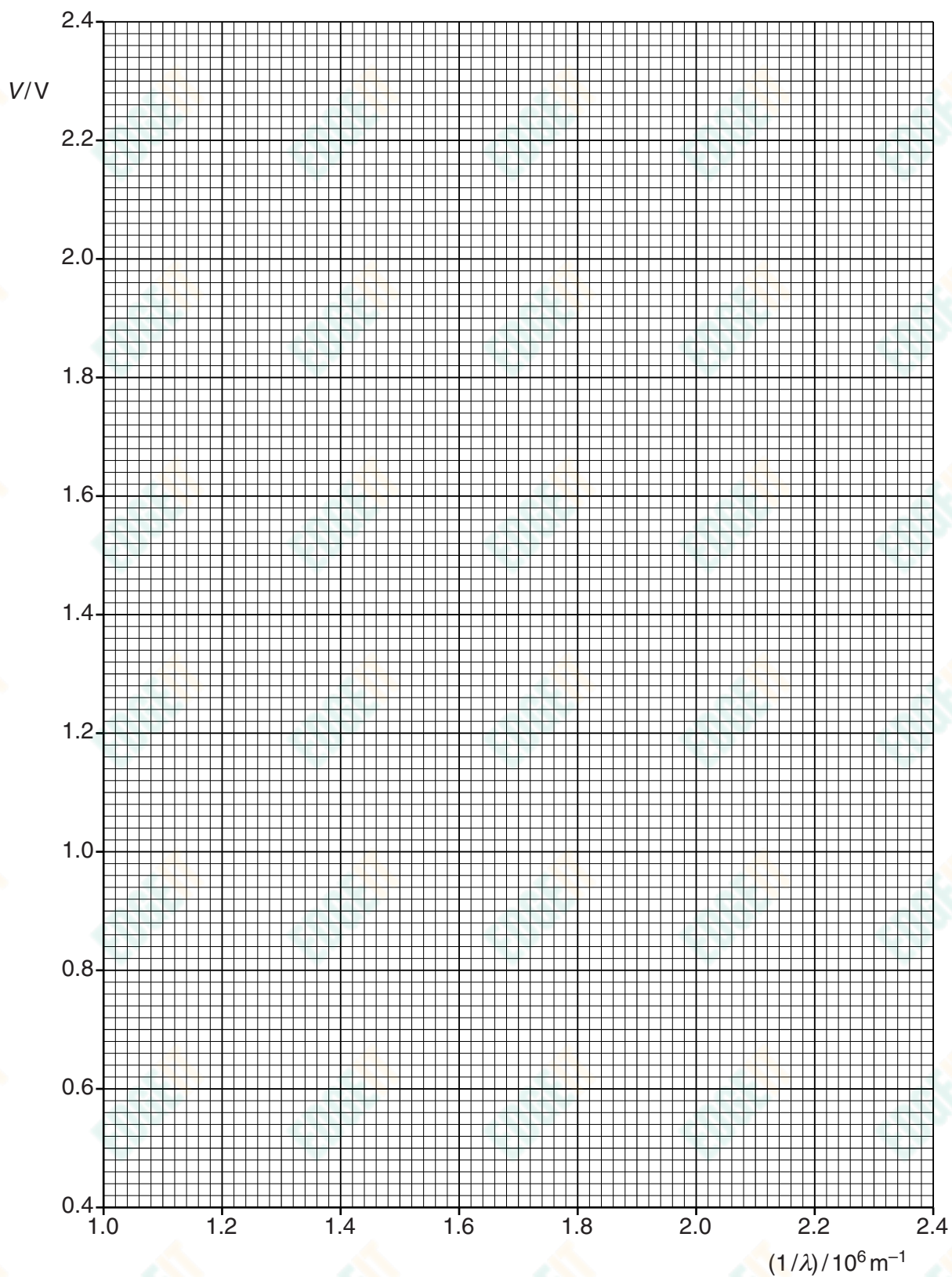
Calculate and record values of $(1/\lambda)/10^6\text{m}^{-1}$ in Fig. 2.2.

[2]

- (c) (i) Plot a graph of V/V against $(1/\lambda)/10^6\text{m}^{-1}$. Include error bars for V . [2]
- (ii) Draw the straight line of best fit and a worst acceptable straight line on your graph. Both lines should be clearly labelled. [2]
- (iii) Determine the gradient of the line of best fit. Include the uncertainty in your answer.

gradient = [2]

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- (iv) Determine the y -intercept of the line of best fit. Include the uncertainty in your answer.

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y -intercept = [2]

- (d) (i) Using your answer to (c)(iii), determine a value for h .

Data: $c = 3.0 \times 10^8 \text{ ms}^{-1}$ and $e = 1.6 \times 10^{-19} \text{ C}$.

$h = \dots\dots\dots \text{ Js}$ [1]

- (ii) Determine the percentage uncertainty in your value of h .

percentage uncertainty = % [1]

- (e) Using your answer to (c)(iv), determine a value for B . Include an appropriate unit and the absolute uncertainty in your answer.

$B = \dots\dots\dots$ [2]

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